

Expt 6	Finite Word length Effects
Date:	6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

Code:

```
clc; clear; close all;

%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0-1)
%% --- FIR High Pass using Hanning Window ---

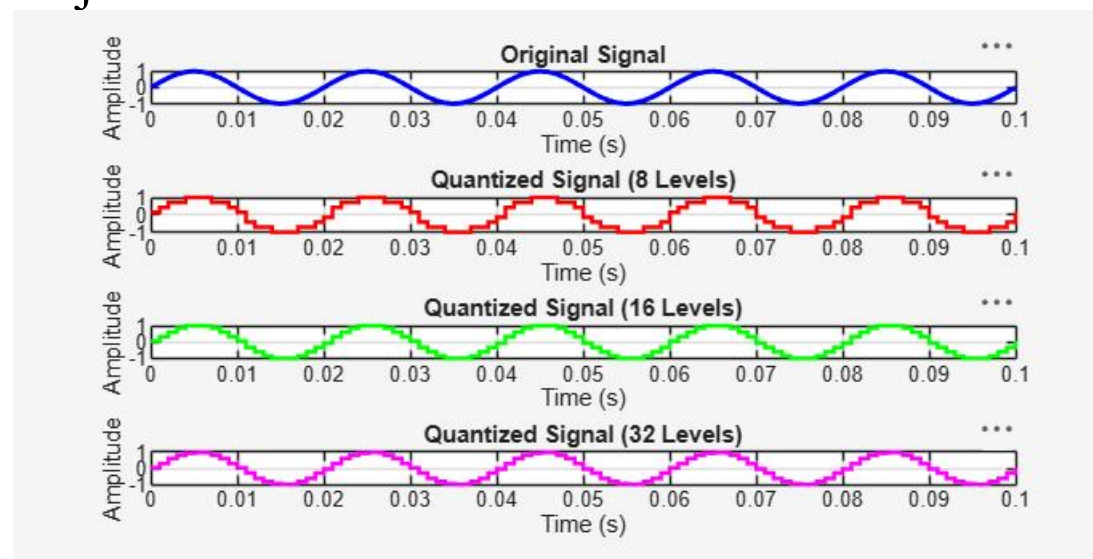
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);
%% --- Frequency Response Plots ---
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

Outputs:

Objective 1:

```
MSE for 8 Levels = 0.010473
MSE for 16 Levels = 0.001763
MSE for 32 Levels = 0.000357
>>
```

Objective 2:



Objective 3:

